

package still requires human review before deployment. AI is therefore not replacing engineers but reducing repetitive workloads and improving speed and decision-making across the embedded lifecycle.

AI integration in embedded systems is supported by a growing ecosystem of tools across requirements, testing, code analysis, and compliance workflows. For requirements analysis, LLMs such as Tulip or Mistral can be used, while tools such as CPPCheck and SonarQube handle static analysis and compliance verification. Test frameworks such as Unity were already widely used, but are now increasingly being integrated into AI-driven workflows.

One major challenge in automotive and embedded development is data security. Many organisations cannot upload source code to cloud-based AI platforms because of compliance restrictions. This is where offline frameworks such as Ollama become useful, allowing LLMs to run locally on internal systems without exposing sensitive data. Ollama also integrates with environments such as VS Code and Linux workflows, enabling AI-assisted pipelines for traceability, compliance-gap detection, documentation, requirements updates, and static-analysis support.

The same concept extends to rule-based AI for architecture generation. Developers can define UML-based design rules and build utilities that generate software architectures aligned with predefined standards. This reduces repetitive effort while improving consistency and compliance coverage.

Deploying AI on microcontrollers adds another layer of complexity because of strict constraints on memory, compute, and power. Depending on the use case, developers may use datasets, training frameworks, pre-trained models, or transfer learning. TinyML frameworks simplify this process with optimised pipelines designed for

- In Design Phase for Embedded System Development
Incorporation of AI can save considerable amount of efforts in development.
- In process adherence repetitive compliance requirement to satisfy need of different work item for process improvement workflow with AI .
- Use of Rule base AI can save design efforts
- Use of LLM Neural Network tools ease the documentation for compliance and process adherence analysis
- Use of Non LLM Neural network and traditional ML model deployment on embedded processor will increase accuracy of system

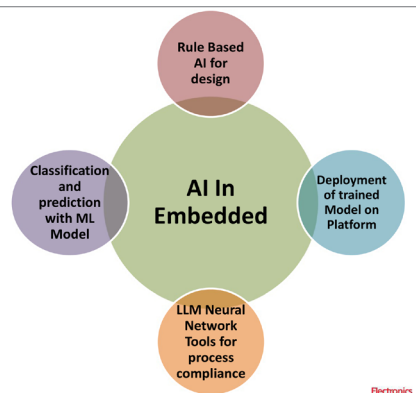


Fig. 3: How AI has improved automotive embedded systems

constrained embedded hardware.

The deployment workflow is hardware-dependent. Models must be quantised and optimised for specific NPUs or microcontroller architectures. Neural-network compilers and inference libraries convert models into MCU-compatible binaries. In most cases, the embedded application remains lightweight and primarily acts as a trigger layer for AI models rather than running full applications on the device.

A simple example is waveform classification, where sensor data is labelled, passed through a model, and classified into different waveform types. This demonstrates how embedded AI can process real-time inputs and generate predictions directly on constrained hardware while operating within embedded limitations.

Intelligent automotive systems horizon

In conclusion, we cannot adopt AI indiscriminately in automotive or embedded systems without adhering to established compliance and validation frameworks. We must always operate within defined safety and governance boundaries. A key framework in this space is ISO 8800, introduced in 2024, which defines how we analyse AI-related risks, select and manage data, ensure data quality, and validate and verify AI workflows. It also defines how we continuously monitor deployed AI

systems, especially in safety-critical environments where behaviour must remain controlled and traceable.

From a takeaway perspective, we can see that LLM-based AI can improve functional safety workflows by accelerating analysis and supporting decision-making, but we must treat it only as a decision-support tool rather than an autonomous process owner. In embedded systems, edge AI running on microcontrollers enables real-time, low-latency execution and reduces cloud dependency, which is essential for fault detection and safety-critical use cases. At the same time, we can use trained models to improve efficiency across design and development workflows, strengthening lifecycle management and compliance alignment.

However, we must also acknowledge that AI outputs are non-deterministic, so we cannot rely on them directly. Human intervention is still required across design, deployment, and validation. We must ensure strong governance, runtime monitoring, and strict adherence to functional safety and cybersecurity standards. Ultimately, we see AI as an enabler that improves efficiency and decision-making, but not as a replacement for human judgement in automotive systems. **EFY**

The article is based on the talk at EFY Expo Pune 2026, titled 'Embedded Design & Process Optimisation Using AI Principles,' featuring a presentation by Sarang Savji, Technical Lead, Varroc Engineering Ltd, India. It has been transcribed and curated by Saba Aafreen, Electronics For You.